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MACHINE LEARNING ARCHITECTURES FOR CROSS-DOMAIN CLASSIFICATION USING TRANSFER LEARNING

Abstract

Certain aspects of the disclosure provide systems and methods for a cross-domain ensemble of machine learning models for onboarding new classification services. A method includes processing a classification task with an ensemble of machine learning models to generate a classification prediction, wherein the ensemble of machine learning comprises a base learner and a meta model, wherein the base learner is trained on a different domain from the classification task. The method further includes training a new classification model based on the classification prediction and data associated with the classification task.

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Background/Summary

BACKGROUND

Field

[0001] Aspects of the present disclosure relate to machine learning, and in particular, to machine learning models for cross-domain classification.

Description of Related Art

[0002] The cold-start problem in machine learning modeling refers to a condition in which a machine learning model is not performant until enough training data has been gathered to train the model sufficiently. The cold start problem may also occur with a trained model when such a model is exposed to a “new” type of input data that is meaningfully different than the data upon which the model was trained.

[0003] Machine learning models are particularly well-suited for providing domain-specific outputs (e.g., inferences) when trained on sufficient domain-specific data. A domain in this context could be, for example, a type of application, business function, or even a specific user. Because it may be technically difficult to train a single model to generalize across many domains, often domain-specific machine learning models are trained and deployed for various useful functions. Training a plurality of domain-specific models generally requires extensive data engineering and model development efforts, which, in-turn, requires significant expenditure of time, processing, and cost resources.

[0004] In some cases, a domain-agnostic (e.g., generalized) model may be used instead of a domain-specific model to avoid the expenditure of time, processing and cost resources, as well as provide a model for immediate use when deploying to a new domain. However, for some critical tasks, a generalized model may not provide sufficient performance (e.g., prediction accuracy). Therefore, in such cases, the cold-start problem remains a technical problem for machine learning for new domains.

[0005] Accordingly, there is a need for improved machine learning modeling techniques for overcoming the cold-start problem and more rapidly deploying performant machine learning models to new domains.

SUMMARY

[0006] Certain aspects provide a computer-implemented method of onboarding a new classification service, comprising: receiving a request for a target classification task, wherein no model in a plurality of machine learning models is trained for the target classification task; selecting one or more trained machine learning models from the plurality of machine learning models for a set of machine learning models; generating a set of classification predictions with the set of machine learning models, wherein each model of the set of machine learning models predicts a respective classification prediction; and processing the set of classification predictions with a meta model trained to generate a label for the target classification task.

[0007] Other aspects provide a computer-implemented method for training a new classification service, comprising: processing a classification input with an ensemble of machine learning models trained to generate a classification label; determining one or more features associated with classification data associated with the classification input; and training a new classification model based on the classification data, the one or more features, and the classification output.

[0008] Other aspects provide processing systems configured to perform the aforementioned methods as well as those described herein; non-transitory, computer-readable media comprising instructions that, when executed by a processors of a processing system, cause the processing system to perform the aforementioned methods as well as those described herein; a computer program product embodied on a computer readable storage medium comprising code for performing the aforementioned methods as well as those further described herein; and a processing system comprising means for performing the aforementioned methods as well as those further described herein.

[0009] The following description and the related drawings set forth in detail certain illustrative features of one or more aspects.

Description

DESCRIPTION OF THE DRAWINGS

[0010] The appended figures depict certain aspects and are therefore not to be considered limiting of the scope of this disclosure.

[0011] FIG. 1 depicts an example system for onboarding a new application with an associated model in an ecosystem including other applications and associated models.

[0012] FIG. 2A depicts an example data flow for utilizing an ecosystem of applications and models, and FIG. 2B depicts an example data flow for onboarding a new application for the ecosystem.

[0013] FIGS. 3A-3C depict example workflows for onboarding a new classification service, including inferencing with a cross-domain ensemble model and training a new classification model for the cross-domain ensemble model.

[0014] FIG. 4 depicts an example method for onboarding a new classification application.

[0015] FIG. 5 depicts an example method for training a new classification model.

[0016] FIG. 6 depicts an example processing system with which aspects of the present disclosure can be performed.

[0017] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the drawings. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0018] Aspects of the present disclosure provide apparatuses, methods, processing systems, and computer-readable mediums for training and inferencing with cross-domain classification models.

[0019] Machine learning models (also referred to as “models” herein) serve many useful functions, including performing various types of classification tasks, which in-turn enable and/or enrich various downstream applications and services. Classification generally involves outputting a prediction (e.g., a class label) based on a given input. Consider the classic example of whether a picture includes an image of a dog or a cat. A classification model may process the input picture and output a predicted class label of “dog” or “cat.” Other examples of classification tasks include recommendation systems for a digital media application, ad-click predictions for an advertising application, spam filtering for an email service, to name just a few.

[0020] Classification-type tasks are abundant in modern electronic systems and processes. For example, in the financial domain, classification of risk is a frequent and important technical function that acts as a gate to many financial transaction types. Generally, when a classification model is to be implemented for a particular task in a given domain, there is some period of time in which the model must be trained on domain-specific data to reach a performance level sufficient to deploy the model. Practically speaking, this means that any system or processes relying on such a model may lack performant functionality when launched, or that the launch may take significantly longer while sufficient data is collected to get the underlying model trained to a sufficient performance level. As above, this is commonly referred to as the cold-start problem.

[0021] It is often the case that an entity (e.g., an organization) seeking to deploy a new model to perform a new classification task has already deployed other models that perform other classification tasks in related domains. By their related domain nature, these existing machine learning models may process certain input data (e.g., features) in common with the new machine learning model. Thus, the existing and new models have at least a subset of common features. In

some examples described herein, a set of related domains may be referred to as an ecosystem.

[0022] Consider as one example, a user ID associated with a user, used for multiple different applications and/or services offered by an organization, may be a common input to existing models and the new model. If the user ID is associated with, for example, a predicted high risk of fraud with respect to one application (as predicted by an associated model), then it follows that the same user ID may indicate risk associated with the user ID for a new application. Thus, as described further herein, the “learning” of existing models (e.g., with respect to the particular user ID) in a related domain may beneficially be transferred to the new machine learning model to more rapidly and efficiently train the new model to a performance level sufficient for deployment. Moreover, the ultimate performance of the new model may exceed that of a model trained without transferring the learning from related domains.

[0023] Further, the outputs of existing models in a related domain may be ensembled and a meta model may be trained to take the outputs of each existing model as inputs in order to generate a further output that shares relevance across the related domains, such as in the risk example above. In such scenarios, the existing models may be referred to as “base models” or “base learners” of the ensemble. The output of the meta model thus benefits from the learning of all the base models as well as its own training. As has been shown in the art, an ensemble of models—even of low performance or “weak” base models—can often perform better than a single model, while also generalizing across domains more readily. Thus, the domain-specific base models plus the meta model create higher performance cross-domain models than a single model trained to generalize across multiple domains.

[0024] Moreover, the output of the meta model (e.g., a predicted classification label) may be used to rapidly generate training data for a new model in a related domain. Thus, a new model may be deployed (or “onboarded”) within an ecosystem of related domain models more quickly and with reduced expenditure of resources.

[0025] Accordingly, aspects described herein overcome the cold start problem through a combination of transfer learning and ensembling existing models in related domains to improve training efficiency and ultimate performance of individual and ensembled models in related domains. These aspects thus represent a technical improvement in the field of machine learning and engender beneficial technical effects compared to conventional techniques, such as reduced training time and reduced resource utilization (e.g., reduced compute, time, and power) when implementing a new model in a related domain. Further, the ensemble models described herein overcome technical challenges associated with single cross-domain models by improving generalization, reducing variance (e.g., sensitive to provided inputs for the learned features), improving task accuracy, and reducing feature noise and bias (e.g., in which a single model relies too heavily on too few features while making a prediction).

Example New Application and Associated Model Onboarding System

[0026] FIG. 1 depicts an example system **100** for onboarding a new application with an associated model in an ecosystem including other applications and associated models.

[0027] In particular, FIG. 1 generally depicts a user **102** that interacts with existing applications **104** and **106**. User **102** may interact with one or more applications through a personal computing device (e.g., a desktop, laptop, smartphone, etc.). In this example, existing applications **104** and **106** are in a related domain ecosystem **110** (e.g., all related to media or financial services offered by an organization). Domains may be related by, for example, product or service type, application type, industry code, input data types, output data types, and the like. Often, applications in related domains will have at least some overlapping features in their input data, and may have outputs of similar type (e.g., a risk prediction). For example, domains may be delineated by input data (e.g., features based on a common schema) for a model (e.g., a base learner), and related domains may include at least a subset of features in common with the domain. Further, in this example, existing applications **104** and **106** process data from and/or associated with user **102** having one or more

common features (e.g., having at least a partially common feature space). By way of example only, common features in the digit media ecosystem may include genera, mood, intensity, and rating. As another example, in the financial transaction ecosystem, common features may include transfer amount, payor identification, payee identification, and routing number.

[0028] Existing application **104** is augmented by a first model **114**, existing application **106** is augmented by second model **116**, and new application **108** will be augmented by new model **118** once new model **118** is trained. A technical function and benefit of system **100**, then, is to train and deploy new model **118** more quickly than by conventional methods.

[0029] In the depicted example, first model **114**, second model **116**, new model **118**, and meta model **112** are collectively an ensemble model and part of an AI component **120**. In some cases, meta model **112** may be referred to as a cross-domain meta model. AI component **120** may augment and enrich the function of applications **104**, **106**, and **108** as well as other applications not depicted in FIG. **1**, such as by way of an inferencing service application programming interface (API), micro service, or similar.

[0030] AI component **120** may implement various types of ensemble model architectures, including bootstrap aggregation (bagging), random forest, sequential decision trees, boosting, adaptive boosting (Adaboost), gradient boosting, extreme gradient boosting (XGBoost), stacking, blending, and others. First model **114**, second model **116**, and new model **118** may comprise base models (e.g., base learners) of the ensemble model architecture to generate respective predictive outputs. Meta model **112** serves as the base model aggregator and can perform aggregate prediction through various mechanisms, such as max voting, averaging, weighted averaging, and others. Meta model **112** utilizes the outputs of the respective base models (e.g., first model **114**, second model **116**, and new model **118**) as input features to generate the aggregated prediction.

[0031] Consider an example in which an organization wishes to deploy new application **108** with personalization features provided by new model **118**. As described herein, the cold-start problem prevents user **102** from immediately receiving personalized output from new application **108** because new model **118** is not yet trained. To overcome this issue, AI component **120** may leverage knowledge learned by first model **114** and second model **116** by way of meta model **112** to generate a useful output for new application **108**. For example, a user's preference may be applicable to applications **104**, **106**, and **108**, and therefore meta model **112** allows user **102** to have an immediate user-specific experience in new application **108** without waiting for new model **118** to be fully trained.

[0032] Further, the outputs of meta model **112** may be used to form training data instance for more rapidly and effectively training new model **118**, as described further with respect to FIGS. **3A-3C**. In this way, transfer learning may be accomplished between the various base models, including first model **114**, second model **116**, and new model **118** in this example. Transfer learning utilizes knowledge learned from a first task to improve performance in a second, related task. Features learned by a model in the first task may be applied to a new model in the second task. The new model may then be trained on new data associated with the second task, while utilizing the pre-trained features. For example, a feature layer of the trained model for the first task may be applied to the new model. As another example, feature layers of the trained model for the first task may be fixed (e.g., weights of features in the fixed layers may not be updated), and new layers are trained based on the second task and data.

[0033] In one example, system **100** may be used for rapidly deploying personalization a personalization model (e.g., new model **118**) in support of a new application (e.g., new application **108**). In particular, system **100** may be configured to facilitate deployment of personalized experiences within new application **108** based on an existing user's (e.g., user **102**) personalized experience in one or more existing applications (e.g., existing applications **104** and **106**) in related domains within related domain ecosystem **110**. For example, a new media recommendation system powered by new model **118** may be based on user **102**'s personalized recommendations in existing

applications **104** and **106** (supported by models **114** and **116**, respectively). Other use cases may include, for example, fraud detection in money movement related domains, image classification in a photography and videography ecosystem, document classification in document processing ecosystem, and the like. In some embodiments, related domains may be determined based on an association between tasks, for example, recommendation tasks, fraud detection, document classification, image/video classification, etc. In some embodiments, related domains may be determined based on a similarity of domains, for example, media domains, image and video domains, document domains, money management domains, etc.

[0034] In another example, system **100** may be used to rapidly make personalized recommendations. Existing first application **104** may be a music application, and the music application may use associated first model **114** to make music recommendations for user **102**. Existing second application **106** may be a television application, and the television application uses associated second model **116** to make television recommendations for user **102**. New application **108** may then be another media application, such a podcast application using associated new model **118** to make podcasts recommendations. Through transfer learning, new model **118** may learn to recommend podcasts based on user **102**'s known preferences for certain types of television shows and music.

[0035] Accordingly, system **100** provides a way to rapidly deploy new applications and associated models to an ecosystem. AI component **120** enables faster training of new model **118** as well as improved prediction performance through meta model **112**.

Example Data Flows for Onboarding New Applications

[0036] FIGS. 2A-2B depict example data flows for using the AI component described herein to onboard a new application with an associated model in an ecosystem including other applications and associated models. In particular, FIG. 2A depicts data flow **200** of payment transaction **202** through an existing application in the ecosystem, for example, existing first application **104** in FIG. 1, utilizing AI component **220** and feature store **230** to output response **212** classifying payment transaction **202**. In this example, the existing application is a payment application **201** associated with a trained base model, such as first model **114** in FIG. 1.

[0037] In this example, payment transaction **202** may be a payment transaction including money movement information, for example, an amount, a date and/or timestamp, internet protocol (IP) address, payee identity, payee routing number, payee device identifier, payor identity, payor routing number, payor device identifier, etc. Such money movement information may include types of data common to a transaction for all related domains in a money movement ecosystem. For example, a payment transaction may include some or all of the same types of data as a payment, a banking transaction, and the like. Thus, payment transaction **202** may be processed by a cross-domain ensemble model **210** associated with the money management ecosystem.

[0038] AI component **220** obtains entity features at step **204** from feature store **230** through featurization service **214**. Such entity features include features associated with the user, e.g., user **102**, and for each domain model (e.g., first model **114** and second model **116** in FIG. 1) of cross-domain ensemble model **210**. In data flow **200**, payment application **201** is augmented by an existing payments model, e.g., first model **114**. Features associated with this existing payments model for the user are stored in payments standardized features store **222**. Featurization service **214** obtains such features from payments standardized features store **222**, to be provided to model execution service **216** for inferencing at step **206**.

[0039] AI component **220** calls for a model inference at step **206** from model execution service **216**. AI component **220** provides entity features obtained at step **204** to model execution service **216** for execution. AI component **220** also provides payment transaction **202** information to model execution service **216**. Model execution service **216** executes on the entity features obtained at step **204** to classify payment transaction **202** using cross-domain ensemble model **210**. For example, model execution service **216** may execute the cross-domain ensemble model **210** associated with

the ecosystem, for example, first model **114** and meta model **112**, using payment transaction **202**. In particular, payment transaction **202** may be provided to first model **114** to generate a fraud risk prediction, and the generated prediction provided to meta model **112** to generate an ensemble classification prediction.

[0040] The classification prediction generated by model execution service **216** is used at step **208** to output the response **212**. Output response **212** may be, for example, a determination based on the classification prediction, a score, and/or the like. In the depicted example, payment transaction **202** may be classified as fraudulent or not fraudulent. For example, output response **212** may include a decision to approve payment transaction **202** and a confidence score.

[0041] FIG. 2B depicts data flow **250** of payroll transaction **252** through a new application **251** in the ecosystem, (e.g., new application **108** in FIG. 1), AI component **220** and feature store **230** to output response **262** classifying payroll transaction **252** as fraudulent or not. In the depicted example, the new application may be a payroll application and an associated user-specific domain model has not been trained, for example, new application **108** and new model **118** described in FIG. 1. However, as described herein, AI component **220** provides a cross-domain ensemble model **210** utilizing related domain models and a cross-domain meta model to generate a user-specific classification of a payroll transaction, even without a trained payroll model.

[0042] Similar to payment transaction **202**, payroll transaction **252** may include money movement information, for example, an amount, a date and/or timestamp, internet protocol (IP) address, payee identity, payee routing number, payee device identifier, payor identity, payor routing number, payor device identifier, etc. Thus, information associated with payroll transaction **252** may be processed by related-domain models, for example, a payments model and a banking model.

[0043] AI component **220** obtains entity features at step **244** from feature store **230** through featurization service **214**. Because payroll is a new application, little to no data on the user has been stored in payroll standardized features **226**. Featurization service **214** uses features associated with the user from other domains in the ecosystem for use in the new domain, to be used by model execution service **216** with the cross-domain ensemble model **210** to generate a user-specific prediction for the new application **251**.

[0044] Generally, features and associated parameter weights from one domain may be useful for determinations in other, related domains. For example, a discriminative feature in one domain may similarly be discriminative in a second domain. Thus, featurization service **214** may obtain common features from other domains for use in the new domain. As an example, features indicating higher risk of fraud may be highly correlated across domains in the money movement ecosystem. Therefore, these features may be used to transfer such learning to a new model for the new domain.

[0045] Featurization service **214** may obtain features from related domains, payments standardized features store **222** and banking standardized features store **224**, to provide to model execution service **216** for inferencing with the cross-domain ensemble model.

[0046] At step **246**, AI component **220** may call for an inference by the cross-domain ensemble model **210** by model execution service **216**. Model execution service **216** generates an inference from the cross-domain ensemble model **210** by providing the obtained features and payroll transaction **252** to the cross-domain ensemble model **210**. Cross-domain ensemble model **210** comprises base learners, e.g., first model **114** and second model **116**, and a meta model, e.g., meta model **112**, associated with a user and an ecosystem. As described in more detail with respect to FIGS. 3A-3C, cross-domain ensemble model **210** utilizes knowledge from the related domains to generate a classification for the new domain. In certain embodiments, a model may be trained for each domain and a meta model may be trained to process the output of each domain-specific model as input, and the meta model outputs a classification prediction. Beneficially, even where there is not a model trained for the new domain, the related domain models and the meta model may be used to generate a classification prediction for the new domain.

[0047] As an example, in a transaction using new application **251**, a device associated with the payor has no historical data in the new payroll domain, but was involved in fraud in another related domain, such as a banking domain. The banking model may indicate a fraudulent transaction, such as with an elevated risk score. The cross-domain ensemble model **210** output would reflect this elevated risk of fraud determined by the banking model. Thus, beneficially, the elevated risk of fraud in the banking domain may be used across the ecosystem, in this example, in the payroll domain even though the payroll domain model may not have historical data to make such a determination. Further, by using an ensemble of domain models and an associated meta model, this cross-domain architecture may be scaled to many related domains.

[0048] At step **248**, a response is generated based on the classification prediction. The output response **262** may be, for example, a determination based on the classification prediction, a score, and/or the like. For example, output response **262** may include a decision to decline the payroll transaction **252** and a confidence score. Further, at step **248**, the response **268** is published to the payroll standardized features **226** in feature store **230**. As described herein, such as with respect to FIG. 3B, payroll standardized features **226** may be used to train a new domain model for payroll to process subsequent payroll transactions.

[0049] Note that FIGS. 2A-2B are just examples of data flows consistent with aspects described herein, but others are possible including additional, alternative, or fewer steps.

Example Flows for Onboarding New Classification Application

[0050] FIGS. 3A-3C depict example workflows for onboarding a new application and training a new model for an example money movement ecosystem. In embodiments, the example workflows are executed by an AI component, for example, AI component **120** in FIG. 1 or AI component **220** in FIG. 2. In this example, an AI component interacts with data store **310** comprising a database associated with each domain, in this example, payment database **304** and banking database **306**. Each domain-specific database may comprise user-specific and domain-specific data and labels. Payroll database **308** is added to data store **310** when the payroll application is onboarded and stores data related to payroll transactions as payroll application is used, as described in further detail herein.

[0051] In this example, the AI component further interacts with feature store **320** configured to store features associated with each domain database, in this example, payment features **314** associated with payment database **304**, and banking features **316** associated with banking database **306**. The AI component includes a set of models **330** comprising a model associated with each domain, in this example, payment model **324** and banking model **326**, which may be examples of first model **114** and second model **116** in FIG. 1. Each model in the set of models **330** is trained to output a classification prediction, in this example, a risk prediction, such as a fraudulent payment transaction, a returned bank transfer, etc. In particular, payment model **324** may generate a first base model prediction **334** and banking model **326** may generate a second base model prediction **336**. Further, each model in the set of models **330** is trained on at least partially common feature space (e.g., having one or more common features). The AI component further includes meta model **342** configured to process the classification predictions from the set of models **330** (e.g., first base model prediction **334** and second base model prediction **336**) and generate meta prediction **344**. Meta model **342** is an example of cross-domain meta model **112** in FIG. 1. Together, the set of models **330** including domain-specific base learners and meta model **342** may comprise an example of cross-domain ensemble model **210** in FIG. 2.

[0052] In particular, FIG. 3A depicts an example workflow **340** for onboarding a new application, for example, the new payroll application. Initially, the new payroll application does not have an associated trained model in the AI component for this user. As described herein, the cold-start problem may prevent the user from immediately receiving a user-specific risk assessment from a new payroll application. However, the new payroll application may leverage the cross-domain ensemble model of the AI component to generate a classification output for payroll transactions in

new payroll application. Trained models associated with classification tasks for similar domains, e.g., base learners for related domains, in this example, payment model **324** and banking model **326**, may be used in the ensemble model to generate this classification output. The cross-domain ensemble model may be beneficially used by the new payroll application to provide the classification output for payroll transactions, even without an associated payroll-specific domain model.

[0053] In the depicted example, workflow **340** overcomes this cold-start problem by processing payroll transaction **302** with the related base learner models. Payment model **324** processes payroll transaction **302** to generate a classification prediction, first base model prediction **334**, which is provided to meta model **342**. First base model prediction **334** may be a risk prediction (e.g., risk of fraud) for payroll transaction **302**. For example, payment model **324** may determine, if the payroll transaction was in the payment domain, whether it would be fraudulent. Beneficially, then, knowledge learned by payment model **324** may be utilized to predict whether payroll transaction is fraudulent.

[0054] Similarly, payroll transaction **302** is processed by banking model **326**. Banking model **326** processes payroll transaction **302** to generate a classification prediction, second base model prediction **336**, and provides the second base model prediction **336** to meta model **342**. Second base model prediction **336** may be a risk prediction (e.g., risk of fraud) for payroll transaction **302**, for example, if payroll transaction **302** were in the banking domain, whether banking model **326** would determine payroll transaction to be fraudulent.

[0055] Meta model **342** is trained to generate meta prediction **344** based on first base model prediction **334** and second base model prediction **336**. Meta prediction **344** may indicate whether payroll transaction **302** is fraudulent based on predictions from related domains, e.g., base learners payment model **324** and banking model **326**. Meta model **342** may be a meta-classifier trained on the outputs of the base learners, for example, first base model prediction **334** and second base model prediction **336** may be the inputs to meta model **342**.

[0056] Meta model **342** may stack or combine the outputs from each of the base learners to determine meta prediction **344**. In some embodiments, meta model **342** may determine meta prediction **344** based on the prediction among the base learners with the most votes, for example, the most frequent prediction among the base learners. In some embodiments, meta model **342** may average the outputs from the base learners, for example, averaging first base model prediction **334** and second base model prediction **336**. In some embodiments, meta model **342** may determine a weighted average, for example, weights of base learners to determine a weighted average. For example, a weight may be applied to each of first base model prediction **334** and second base model prediction **336** to determine meta prediction **344**. Generally, meta model **342** may have improved performance as compared to any individual model because it may minimize or reduce error in one base learner through the combination of predictions from many base learners.

[0057] Furthermore, meta model **342** may provide improved resilience as compared to a single domain-specific model because meta model **342** may not be overly reliant on a specific base learner. If a specific base learner were to have reduced performance, for example, it stopped being called or provides bad data, etc., such fragility may be reduced because meta model **342** also utilizes other base learners for prediction.

[0058] Although depicted in this example as two base learners, e.g., payment model **324** and banking model **326**, in some embodiments, meta model **342** may utilize many base learner models, for example, about 100 base learners, greater than 100 base learners, about 1000 base learners, greater than 1000 base learners, etc. In some embodiments, a large number of base learners may result in poor scalability of meta model **342**, and meta model **342** may be restricted to a maximum number of base learners during training of meta model **342**.

[0059] Thus, beneficially, meta prediction **344** is generated for payroll transaction **302**, even without a domain-specific trained model through the ensembling of the base learners for related

domains and the meta model **342**. Further, data associated with payroll transaction **302** is stored in payroll database **308** to facilitate later training, such as in workflow **350** of FIG. **3B**.

[0060] The base learners in the set of models **330** may be identified based on related domains, similarity in target tasks, and/or association with meta model **342**. In some embodiments, many base learners are trained for a variety of classification tasks and base models may be selected for inclusion in the set of models **330** based on a similarity between a base learner's respective classification task and the target classification task associated with the new application. In some embodiments, the new application is associated with a target domain, and each base learner selected for inclusion in the set of models **330** based on a similarity between the target domain and a domain of the base learner, e.g., based on at least a subset of common features. In some embodiments, base learners are selected for inclusion in the set of models **330** based on an association between the base learner and the meta model, for example, a pre-existing ensemble model. The ensemble model (including the meta model and the set of base learners) may be selected based on a similarity of the ensemble model and the new application, e.g., related domains, similar classification tasks, common data, etc.

[0061] FIG. **3B** depicts an example workflow **350** for training a model for the new application as it is utilized. Over time, payroll data is accumulated and stored in payroll database **308**. This data is used to generate and train a new base learner for the payroll domain, payroll model **328**. Thus, payroll model **328** may be rapidly and effectively trained.

[0062] In particular, a subsequent payroll transaction, second payroll transaction **352** for the new payroll application is processed, such as described with respect to FIG. **3A** to generate an associated payroll prediction, meta prediction **364** from meta model **342**. Payroll data associated with second payroll transaction **352** is stored in payroll database **308**. Such data may be associated with meta prediction **364** to generate training data for training payroll model **328**. For example, second payroll transaction **352** may be processed by payments model **324** and banking model **326**, and the outputs of each base learner processed by meta model **342** to generate meta prediction **364**, such as described with respect to FIG. **3A**, for example, a fraudulent transaction prediction. Payroll model **328** may be instantiated and trained based on model training data **366** comprising second payroll transaction **352** and meta prediction **364** to generate a fraudulent transaction prediction for payroll transactions.

[0063] Further, each base learner in the ecosystem is trained based on a common feature space, and the new base learner, e.g., payroll model **328**, utilizes such common features. Knowledge about common features may be transferred from related base learners. As an example, payroll model **328** may beneficially be further trained based on one or more model features learned in a related domain base learner through transfer learning. In some embodiments, one or more model feature weights from a related domain base learner, e.g., payments model **324** and/or banking model **326**, may be used to determine one or more model feature weights of payroll model **328**. By using the trained features, less additional training and added training data may be used to generate a mature model. Thus, beneficially, payroll model **328** may be trained faster, on less data, and/or with higher performance, as compared to a model trained without transfer learning.

[0064] Further, once payroll model **328** has been trained and generated, payroll model **328** may be added to the set of models **330** and used by meta model **342**. Meta model **342** may then utilize payroll model **328** as a base learner, along with payment model **324** and banking model **326**.

[0065] FIG. **3C** depicts an example workflow **360** for inferencing with new payroll model **328**, generated and trained as described with respect to FIG. **3B**, as part of meta model **342**. Once new payroll model **328** has matured (e.g., been fully trained as described with respect to FIG. **3B**), added to the set of models **330**, and ensembled with meta model **342**, payroll model **328** may be used to generate meta prediction **374** for payroll transaction **362**.

[0066] As depicted, payroll transaction **362** is a subsequent payroll transaction after payroll model **328** has been trained. Payroll transaction **362** is processed by a set of models **330** associated with

the payroll domain and associated domains, in this example, payments and banking.

[0067] Payroll transaction **362** is processed by payment model **324** to generate first base model prediction **334**, in this example, predicting whether payroll transaction **362** is fraudulent in a payment domain.

[0068] Similarly, payroll transaction **362** is processed by banking model **326** to generate second base model prediction **336**, in this example, predicting whether payroll transaction **362** is fraudulent in a payment domain.

[0069] Additionally, payroll transaction **362** is processed by payroll model **328** to generate third base model prediction **338**, in this example, predicting whether payroll transaction **362** is fraudulent. Payroll model **328** processes payroll transaction **362**, as well as payroll data stored in payroll database **308** based on payroll features **318** to generate third base model prediction **338**.

[0070] Meta model **342** generates a meta prediction **374** based on the first base model prediction **334**, second base model prediction **336**, and third base model prediction **338**. Thus, meta prediction **374** is based on domain-specific data and a domain-specific model, e.g., payroll model **328**.

Further, beneficially, meta prediction **374** is also based on data and models from related domains, e.g., payments and banking. These other related domains may provide beneficial relevance when assessing the payroll transaction **362**. Further, meta model **342** ensembles the set of machine learning models to provide improved performance over just one base learner, e.g., payroll model **328**.

[0071] As an example, an individual entity involved in a transaction may have no history in the target domain, e.g., payroll, but was involved in fraud in a different related domain, such as banking. Based on the banking data in banking database **306**, banking model **326** may determine a higher risk of fraud for the payroll transaction in the banking model. Ultimately, meta prediction **374** may have a higher risk of fraud, based in part on the banking prediction, e.g., second base model prediction **336** used by meta model **342**. Thus, elevated risk in a related domain is captured by meta prediction **374**.

[0072] In some embodiments, a domain may include additional features, for example, in the payments domain, a payment transaction includes a device identifier and in the banking domain a banking transaction does not include a device identifier. When the payments base learner is trained with a feature vector including a device identifier, payments transactions will have a device identifier and the payments model expects this device identifier. However, when the payments base learner processes a banking transaction without a device identifier, the model may behave erratically because of the missing feature, and thus reduce utility of a banking meta model. In some embodiments, entity sparsity enhancement may be used to nullifying sets of features related to entities in some observations during model training. Entity sparsity enhancement may be used, even after maturity, to ensure each base learner model is more robust to other input data from other, new domains as each ensemble model gains more base learners. Entity sparsity enhancement may be used to null features, such as a device identifier, to train base learners to be robust to such sparsity between domains. In some embodiments, the set of features may be nullified based on a relation between each of the features, such that all the features in the set of features may be expected to be null (e.g., missing) in the same transaction, for example, a payment transaction may generally include bank account features, for example, a bank account number, a routing number, a SWIFT code, etc. During training of the payment model, such bank account features may be nullified to ensure the payment model is trained to accommodate inputs from other domains, where such inputs may not provide such bank account features.

[0073] Note that FIGS. 3A-3C are just examples of data flows consistent with aspects described herein, but others are possible including additional, alternative, or fewer steps.

Example Method for Onboarding a New Classification Application

[0074] FIG. 4 depicts an example method **400** onboarding a new classification application, for example, a new application **108** in FIG. 1, or a new payroll application as described with respect to

FIGS. 2A-2B, and FIGS. 3A-3C.

[0075] Initially, method **400** begins at step **402** with receiving a request for a target classification task, wherein no model in a plurality of machine learning models is trained for the target classification task, such as described with respect to FIGS. 2A and 3A. For example, a new classification task associated with the new application and there is not an existing pre-trained model for that domain-specific task.

[0076] Method **400** proceeds to step **404** with selecting one or more trained machine learning models from the plurality of machine learning models for a set of machine learning models. In some embodiments, each model of the set of machine learning models is trained based on a common feature space, and the target classification task is associated with the common feature space, for example, as described with respect to FIGS. 2A-2B. For example, the set of machine learning models in an ecosystem of related domains.

[0077] In some embodiments, each respective model of the plurality of machine learning models is trained for a respective classification task, and the one or more trained machine learning models are selected from the plurality of machine learning models based on an association between the respective classification task and the target classification task.

[0078] In some embodiments, the target classification task is associated with a target domain, each respective model in the plurality of machine learning models is trained on data associated with a respective domain, and the one or more trained machine learning models are selected from the plurality of machine learning models based on a similarity between the respective domain and the target domain.

[0079] In some embodiments, the set of machine learning models is selected based on an association of each respective model in the set of machine learning models with a meta model, and the meta model is selected based on a similarity with the target domain, e.g., related domains, common feature space, respective classification task, etc.

[0080] Method **400** then proceeds to step **406** with generating a set of classification predictions with the set of machine learning models, wherein each model of the set of machine learning models predicts a respective classification prediction, for example, first base model prediction **334** and second base model prediction **336** in FIGS. 3A-3C.

[0081] Method **400** then proceeds to step **408** with processing the set of classification predictions with a meta model trained to generate a label for the target classification task, such as meta model **342** in FIGS. 3A-3C to generate a meta prediction.

[0082] In some embodiments, method **400** further comprises training a new model to generate the label for the target classification task, for example, as described with respect to FIG. 3B. In some embodiments, method **400** further comprises adding the trained new model to the set of machine learning models, e.g., for use with a cross-domain ensemble model.

[0083] In some embodiments, method **400** further comprises receiving a second request for the target classification task; and generating a second label for the target classification task by processing the second request with the trained new model, for example, as described with respect to FIG. 3C e.g., using the cross-domain ensemble model.

[0084] Note that FIG. 4 is just one example of a method consistent with aspects described herein, but others are possible including additional, alternative, or fewer steps.

Example Method for Training a New Classification Application

[0085] FIG. 5 depicts an example method **500** for training a new classification application, for example, a new application **108** in FIG. 1, or a new payroll application as described with respect to FIGS. 3B.

[0086] Initially, method **500** begins at step **502** with processing a classification input with an ensemble of machine learning models trained to generate a classification label. In some embodiments, the ensemble of machine learning models comprises a pre-trained classification model and a meta model, for example as described with respect to payment model **324**, banking

model **326**, and meta model **342** in FIGS. **3A-3C**.

[0087] Method **500** proceeds to step **504** with obtaining classification data associated with the classification input, for example, as described with respect to payroll database **308** in in FIGS. **3A-3C**.

[0088] Method **500** then proceeds to step **506** with determining one or more features associated with the classification data for example as described with respect to payroll features **318** in FIGS. **3A-3C**. In some embodiments, features may be obtained using transfer learning, for example, one or more features and parameter weights, or frozen feature layers.

[0089] Method **500** then proceeds to step **508** with training a new classification model based on the classification data, the one or more features, and the classification output, for example, as described with respect to payroll model **328** in FIG. **3B**.

[0090] In some embodiments, the pre-trained classification model is associated with a first domain; the new classification model is associated with a second domain; and the classification input is associated with the second domain. In some embodiments, the pre-trained classification model and the new classification model are trained on a common feature space.

[0091] In some embodiments, method **500** further comprises adding the new classification model to the ensemble of machine learning models, e.g., the cross-domain ensemble model **210** in FIG. **2B**.

[0092] Note that FIG. **5** is just one example of a method consistent with aspects described herein, but others are possible including additional, alternative, or fewer steps.

Example Processing System for Onboarding New Applications

[0093] FIG. **6** depicts an example processing system **600** configured to perform various aspects described herein, including, for example, data flows for onboarding new applications as described above with respect to FIGS. **2A-2B**, flows for onboarding new classification applications as described with respect to FIGS. **3A-3C**, method **400** for onboarding a new classification application as described above with respect to FIG. **4**, or method **500** for training a new classification application as described above with respect to FIG. **5**.

[0094] Processing system **600** is generally be an example of an electronic device configured to execute computer-executable instructions, such as those derived from compiled computer code, including without limitation personal computers, tablet computers, servers, smart phones, smart devices, wearable devices, augmented and/or virtual reality devices, and others.

[0095] In the depicted example, processing system **600** includes one or more processors **602**, one or more input/output devices **604**, one or more display devices **606**, one or more network interfaces **608** through which processing system **600** is connected to one or more networks (e.g., a local network, an intranet, the Internet, or any other group of processing systems communicatively connected to each other), and computer-readable medium **612**. In the depicted example, the aforementioned components are coupled by a bus **610**, which may generally be configured for data exchange amongst the components. Bus **610** may be representative of multiple buses, while only one is depicted for simplicity.

[0096] Processor(s) **602** are generally configured to retrieve and execute instructions stored in one or more memories, including local memories like computer-readable medium **612**, as well as remote memories and data stores. Similarly, processor(s) **602** are configured to store application data residing in local memories like the computer-readable medium **612**, as well as remote memories and data stores. More generally, bus **610** is configured to transmit programming instructions and application data among the processor(s) **602**, display device(s) **606**, network interface(s) **608**, and/or computer-readable medium **612**. In certain embodiments, processor(s) **602** are representative of one or more central processing units (CPUs), graphics processing unit (GPUs), tensor processing unit (TPUs), accelerators, and other processing devices.

[0097] Input/output device(s) **604** may include any device, mechanism, system, interactive display, and/or various other hardware and software components for communicating information between

processing system **600** and a user of processing system **600**. For example, input/output device(s) **604** may include input hardware, such as a keyboard, touch screen, button, microphone, speaker, and/or other device for receiving inputs from the user and sending outputs to the user.

[0098] Display device(s) **606** may generally include any sort of device configured to display data, information, graphics, user interface elements, and the like to a user. For example, display device(s) **606** may include internal and external displays such as an internal display of a tablet computer or an external display for a server computer or a projector. Display device(s) **606** may further include displays for devices, such as augmented, virtual, and/or extended reality devices. In various embodiments, display device(s) **606** may be configured to display a graphical user interface.

[0099] Network interface(s) **608** provide processing system **600** with access to external networks and thereby to external processing systems. Network interface(s) **608** can generally be any hardware and/or software capable of transmitting and/or receiving data via a wired or wireless network connection. Accordingly, network interface(s) **608** can include a communication transceiver for sending and/or receiving any wired and/or wireless communication.

[0100] Computer-readable medium **612** may be a volatile memory, such as a random access memory (RAM), or a nonvolatile memory, such as nonvolatile random access memory (NVRAM), or the like. In this example, computer-readable medium **612** includes modelling component **614**, featurization component **616**, data store component **618**, model data **620**, feature store **622**, and domain-specific databases **624**.

[0101] In certain embodiments, modelling component **614** is configured to generate, train, and execute one or more machine learning models of a plurality of machine learning models storing model data **620**, for example, for the set of machine learning models and the meta model in FIG. 4, or the ensemble of machine learning models in FIG. 5. In embodiments, modelling component **614** is further configured to train a new classification model, for example as described with respect to FIG. 4 or FIG. 5. In embodiments, modelling component **614** may be an example of an AI component, for example as described with respect to FIGS. 1-3C.

[0102] In certain embodiments, featurization component **616** is configured to determine and obtain features associated with machine learning models stored in model data **620**. Featurization component **616** may be further configured to store features in feature store **622**, such as one or more domain-specific feature databases in feature store **230** in FIGS. 2A-2B.

[0103] In certain embodiments, data store component **618** is configured to store and retrieve domain data stored in one or more domain-specific databases **624**, such as payment database **304**, banking database **306**, or payroll database **308**, as described with respect to FIGS. 3A-3C.

[0104] Note that FIG. 6 is just one example of a processing system consistent with aspects described herein, and other processing systems having additional, alternative, or fewer components are possible consistent with this disclosure.

Example Clauses

[0105] Implementation examples are described in the following numbered clauses:

[0106] Clause 1: A computer-implemented method for onboarding a new classification application, comprising: receiving a request for a target classification task, wherein no model in a plurality of machine learning models is trained for the target classification task; selecting one or more trained machine learning models from the plurality of machine learning models for a set of machine learning models; generating a set of classification predictions with the set of machine learning models, wherein each model of the set of machine learning models predicts a respective classification prediction; and processing the set of classification predictions with a meta model trained to generate a label for the target classification task.

[0107] Clause 2: The computer-implemented method of Clause 1, further comprising training a new model to generate the label for the target classification task.

[0108] Clause 3: The computer-implemented method of Clause 2, further comprising: receiving a second request for the target classification task; and generating a second label for the target

classification task by processing the second request with the trained new model.

[0109] Clause 4: The computer-implemented method of any one of Clauses 2-3, further comprising adding the trained new model to the set of machine learning models.

[0110] Clause 5: The computer-implemented method of Clause 4, further comprising: receiving a second request for the target classification task; and generating a second label for the target classification task, comprising: generating a second set of classification predictions with the set of machine learning models and the trained new model; and processing the second set of classification predictions with the meta model trained to generate a label for the second request.

[0111] Clause 6: The computer-implemented method of any one of Clauses 1-5, wherein: each model of the set of machine learning models is trained based on a common feature space; and the target classification task is associated with the common feature space.

[0112] Clause 7: The computer-implemented method of any one of Clauses 1-6, wherein the target classification task is associated with the common feature space.

[0113] Clause 8: The computer-implemented method of any one of Clauses 1-7, wherein: each respective model of the plurality of machine learning models is trained for a respective classification task, and the one or more trained machine learning models are selected from the plurality of machine learning models based on an association between the respective classification task and the target classification task.

[0114] Clause 9: The computer-implemented method of any one of Clauses 1-8, wherein: the target classification task is associated with a target domain, each respective model in the plurality of machine learning models is trained on data associated with a respective domain, and the one or more trained machine learning models are selected from the plurality of machine learning models based on an similarity between the respective domain and the target domain.

[0115] Clause 10: The computer-implemented method of any one of Clauses 1-9, wherein the set of machine learning models and the Meta model comprise a cross-domain ensemble model.

[0116] Clause 11: A computer-implemented method for training a new classification application, comprising: processing a classification input with an ensemble of machine learning models trained to generate a classification label; obtaining classification data associated with the classification input; determining one or more features associated with the classification data; and training a new classification model based on the classification data, the one or more features, and the classification output.

[0117] Clause 12: The computer-implemented method of Clause 11, further comprising adding the new classification model to the ensemble of machine learning models.

[0118] Clause 13: The computer-implemented method of any one of Clauses 11-12, wherein the ensemble of machine learning models comprises a pre-trained classification model and a meta model.

[0119] Clause 14: The computer-implemented method of Clause 13, wherein: the pre-trained classification model is associated with a first domain; the new classification model is associated with a second domain; and the classification input is associated with the second domain.

[0120] Clause 15: The computer-implemented method of any one of Clauses 13-14, wherein the pre-trained classification model and the new classification model are trained on a common feature space.

[0121] Clause 16: A processing system, comprising: a memory comprising computer-executable instructions; and a processor configured to execute the computer-executable instructions and cause the processing system to perform a method in accordance with any one of Clauses 1-15.

[0122] Clause 17: A processing system, comprising means for performing a method in accordance with any one of Clauses 1-15.

[0123] Clause 18: A non-transitory computer-readable medium storing program code for causing a processing system to perform the steps of any one of Clauses 1-15.

[0124] Clause 19: A computer program product embodied on a computer-readable storage medium

comprising code for performing a method in accordance with any one of Clauses 1-15.

Additional Considerations

[0125] The preceding description is provided to enable any person skilled in the art to practice the various embodiments described herein. The examples discussed herein are not limiting of the scope, applicability, or embodiments set forth in the claims. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments. For example, changes may be made in the function and arrangement of elements discussed without departing from the scope of the disclosure. Various examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various steps may be added, omitted, or combined. Also, features described with respect to some examples may be combined in some other examples. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method that is practiced using other structure, functionality, or structure and functionality in addition to, or other than, the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

[0126] As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c or any other ordering of a, b, and c).

[0127] As used herein, the term “determining” encompasses a wide variety of actions. For example, “determining” may include calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), ascertaining and the like. Also, “determining” may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory) and the like. Also, “determining” may include resolving, selecting, choosing, establishing and the like.

[0128] The methods disclosed herein comprise one or more steps or actions for achieving the methods. The method steps and/or actions may be interchanged with one another without departing from the scope of the claims. In other words, unless a specific order of steps or actions is specified, the order and/or use of specific steps and/or actions may be modified without departing from the scope of the claims. Further, the various operations of methods described above may be performed by any suitable means capable of performing the corresponding functions. The means may include various hardware and/or software component(s) and/or module(s), including, but not limited to a circuit, an application specific integrated circuit (ASIC), or processor. Generally, where there are operations illustrated in figures, those operations may have corresponding counterpart means-plus-function components with similar numbering.

[0129] The following claims are not intended to be limited to the embodiments shown herein, but are to be accorded the full scope consistent with the language of the claims. Within a claim, reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. No claim element is to be construed under the provisions of 35 U.S.C. § 112 (f) unless the element is expressly recited using the phrase “means for” or, in the case of a method claim, the element is recited using the phrase “step for.” All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

Claims

1. A computer-implemented method for onboarding a new classification service, comprising: receiving a request for a target classification task, wherein no model in a plurality of machine learning models is trained for the target classification task; selecting one or more trained machine learning models from the plurality of machine learning models for a set of machine learning models; generating a set of classification predictions with the set of machine learning models, wherein each model of the set of machine learning models predicts a respective classification prediction; and processing the set of classification predictions with a meta model trained to generate a label for the target classification task.
2. The computer-implemented method of claim 1, further comprising training a new model to generate the label for the target classification task.
3. The computer-implemented method of claim 2, further comprising: receiving a second request for the target classification task; and generating a second label for the target classification task by processing the second request with the trained new model.
4. The computer-implemented method of claim 2, further comprising adding the trained new model to the set of machine learning models.
5. The computer-implemented method of claim 4, further comprising: receiving a second request for the target classification task; and generating a second label for the target classification task, comprising: generating a second set of classification predictions with the set of machine learning models and the trained new model; and processing the second set of classification predictions with the meta model trained to generate a label for the second request.
6. The computer-implemented method of claim 1, wherein: each model of the set of machine learning models is trained based on a common feature space; and the target classification task is associated with the common feature space.
7. The computer-implemented method of claim 1, wherein: each respective model of the plurality of machine learning models is trained for a respective classification task, and the one or more trained machine learning models are selected from the plurality of machine learning models based on an association between the respective classification task and the target classification task.
8. The computer-implemented method of claim 1, wherein: the target classification task is associated with a target domain, each respective model in the plurality of machine learning models is trained on data associated with a respective domain, and the one or more trained machine learning models are selected from the plurality of machine learning models based on a similarity between the respective domain and the target domain.
9. The computer-implemented method of claim 1, wherein the set of machine learning models and the meta model comprise a cross-domain ensemble model.
10. A computer-implemented method for training a new classification service, comprising: processing a classification input with an ensemble of machine learning models trained to generate a classification label; determining one or more features associated with classification data associated with the classification input; and training a new classification model based on the classification data, the one or more features, and the classification label.
11. The computer-implemented method of claim 10, further comprising adding the new classification model to the ensemble of machine learning models.
12. The computer-implemented method of claim 10, wherein the ensemble of machine learning models comprises a pre-trained classification model and a meta model.
13. The computer-implemented method of claim 12, wherein: the pre-trained classification model is associated with a first domain; and the new classification model is associated with a second domain.
14. The computer-implemented method of claim 12, wherein the pre-trained classification model

and the new classification model are trained on a common feature space.

15. A processing system, comprising: a memory comprising computer-executable instructions; and a processor configured to execute the computer-executable instructions and cause the processing system to: receive a request for a target classification task, wherein no model in a plurality of machine learning models is trained for the target classification task; select one or more trained machine learning models from the plurality of machine learning models for a set of machine learning models; generate a set of classification predictions with the set of machine learning models, wherein each model of the set of machine learning models predicts a respective classification prediction; and process the set of classification predictions with a meta model trained to generate a label for the target classification task.

16. The processing system of claim 15, wherein the processor is further configured to cause the processing system to train a new model to generate the label for the target classification task.

17. The processing system of claim 16, wherein the processor is further configured to cause the processing system to: receive a second request for the target classification task; and generate a second label for the target classification task by processing the second request with the trained new model.

18. The processing system of claim 16, wherein the processor is further configured to cause the processing system to add the trained new model to the set of machine learning models.

19. The processing system of claim 18, wherein the processor is further configured to cause the processing system to: receive a second request for the target classification task; and generate a second label for the target classification task, comprising: generate a second set of classification predictions with the set of machine learning models and the trained new model; and process the second set of classification predictions with the meta model trained to generate a label for the second request.

20. The processing system of claim 15, wherein: the target classification task is associated with a target domain, each respective model in the plurality of machine learning models is trained on data associated with a respective domain, and the one or more trained machine learning models are selected from the plurality of machine learning models based on a similarity between the respective domain and the target domain.
